

Tutorial Questions (Set-I)

Q1 Let $<$ be a strict order on a field $\mathbb{K}$, i.e., $(\mathbb{K}, <)$ be an ordered field. Consider the subset $\mathbf{P} = \{x \in \mathbb{K} : x > 0\}$ of positive elements of $\mathbb{K}$. Then show that

- (i) $x, y \in \mathbf{P} \Rightarrow x + y \in \mathbf{P}$ and $xy \in \mathbf{P}$.
- (ii) For all $x \in \mathbb{K}$, one and only one of the following hold :

$$x \in \mathbf{P}, \quad \text{or} \quad x = 0, \quad \text{or} \quad -x \in \mathbf{P}.$$

On the other hand, if $\mathbb{K}$ is a field and $\mathbf{P}$ is a subset that satisfies property (i) and (ii), then show that there is a strict order $<$ on the field $\mathbb{K}$ given by $x < y$ if $y - x \in \mathbf{P}$. Further, show that $(\mathbb{K}, <)$ is an ordered field and $\mathbf{P} = \mathbb{K}^+$ is the subset of positive elements of $\mathbb{K}$.

Q2 Let $(\mathbb{K}, <)$ be an ordered field. Show that

- (a) If $x \neq 0$ in $\mathbb{K}$, then $x^2 > 0$.
- (b) If $x, y \in \mathbb{K}$ with $0 < x < y$, then $0 < y^{-1} < x^{-1}$.
- (c) The field $\mathbb{Q}$ of rational numbers is isomorphic to an ordered subfield of $\mathbb{K}$.

Q3 A field isomorphism $f: \mathbb{K} \rightarrow \mathbb{K}$ is called an *automorphism* of $\mathbb{K}$.

- (a) Show that an automorphism of $\mathbb{Q}$ (or $\mathbb{R}$) must be the identity automorphism.
- (b) What are automorphism of $\mathbb{Q}[\sqrt{2}]$? What are automorphisms $f: \mathbb{C} \rightarrow \mathbb{C}$ such that $f|_{\mathbb{R}}$ is the identity automorphism of $\mathbb{R}$?

Q4 Show that the field $\mathbb{R}(x)$ of rational functions over $\mathbb{R}$ is an ordered field. Show that $\mathbb{R}(x)$ is not Dedekind complete. Also, show that $\mathbb{R}(x)$ is non-Archimedean.

Q5 Let $(a_n)_{n \in \mathbb{N}}$ be a convergent sequence of real numbers.

- (a) Show that $(a_n)_{n \in \mathbb{N}}$ is bounded and a Cauchy sequence.
- (b) Show that every subsequence of a convergent sequence is convergent and has the same limit.

Q6 Define $\limsup_{n \rightarrow \infty} a_n$ and $\liminf_{n \rightarrow \infty} a_n$ of a real sequence $(a_n)_{n \in \mathbb{N}}$. Show that

$$\liminf_{n \rightarrow \infty} a_n \leq \limsup_{n \rightarrow \infty} a_n.$$

Also, prove that $(a_n)_{n \in \mathbb{N}}$ is convergent if and only if $\liminf_{n \rightarrow \infty} a_n = \limsup_{n \rightarrow \infty} a_n$

Q7 Let $(x_n)_{n \in \mathbb{N}}$ be a Cauchy sequence of real numbers. Show that

- (a) $(x_n)_{n \in \mathbb{N}}$ is bounded.
- (b) $(x_n)_{n \in \mathbb{N}}$ has a convergent subsequence.
- (c) Every Cauchy sequence of real numbers is convergent, i.e., the field $\mathbb{R}$ of real numbers is Cauchy complete.

Q8 Let $b > 1$ be a real number. If $r = \frac{p}{q}$, $q \in \mathbb{N}$, and $p \in \mathbb{Z}$, then define $b^r = (b^p)^{\frac{1}{q}}$. Show that b^r is well-defined and $b^{r+s} = (b^r)(b^s)$ for $r, s \in \mathbb{Q}$. If $x \in \mathbb{R}$, then set

$$B(x) = \{b^t : t \in \mathbb{Q} \text{ with } t \leq x\}.$$

Show that $b^r = \sup B(r)$ for $r \in \mathbb{Q}$. Define $b^x = \sup B(x)$ for $x \in \mathbb{R}$. Show that $b^{x+y} = (b^x)(b^y)$ for $x, y \in \mathbb{R}$.

Q9 (**Optional**) Let $(\mathbb{K}, <)$ be an ordered field.

- (a) Show that $(\mathbb{K}, <)$ is Dedekind complete if and only if $(\mathbb{K}, <)$ is Archimedean and Cauchy complete.
- (b) Let $(\mathbb{K}, <)$ and $(\mathbb{K}', <')$ be complete ordered fields. Show that there is a unique isomorphism $f: \mathbb{K} \rightarrow \mathbb{K}'$, which is order-preserving.

Q10 (**Optional**) [**Cantor's construction of real numbers**] Let $\mathcal{C}(\mathbb{Q})$ be the set of $\mathbb{Q}$ -Cauchy sequences i.e., Cauchy sequences of rational numbers.

- (a) Show that $\mathcal{C}(\mathbb{Q})$ is a commutative ring with identity.
- (b) Show that $\mathcal{I} = \{(a_n) \in \mathcal{C}(\mathbb{Q}) : \lim_{n \rightarrow \infty} a_n = 0\}$ is an ideal of $\mathcal{C}(\mathbb{Q})$.
- (c) Show that the quotient ring $\mathcal{R} = \frac{\mathcal{C}(\mathbb{Q})}{\mathcal{I}}$ is a field.
- (d) Define a strict order $<$ on $\mathcal{R}$ such that $(\mathcal{R}, <)$ is an ordered field.
- (e) Show that the field $(\mathcal{R}, <)$ is Dedekind complete.